

Comparing Three-Digit Numbers

Answer Key

Grade 2 Math

Quick Answers

- | | |
|-------------------------|-------------------------|
| 1. $>$ | 6. $>$ |
| 2. $<$ | 7. 852 |
| 3. $>$ | 8. 313, 318, 381 |
| 4. comparison = $<$, — | 9. comparison = $<$, — |
| 5. 251, 512, 521 | 10. 709, 790, 907, 970 |
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What is verified

8 of 10 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 4, 9. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 2 Math
2.NBT.A – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–4 of 5 across 12 tagged checks.

Common wrong answers

7. *If they got 258: made the smallest number instead of the greatest*

- Write $<$, $>$, or $=$: 372 vs. 327
Solution: Hundreds tie (3 and 3), so compare tens: 7 tens beat 2 tens. $372 > 327$
- Write $<$, $>$, or $=$: 458 vs. 485
Solution: Hundreds tie (4 and 4); tens decide: 5 tens are fewer than 8 tens. $458 < 485$
- Compare the expanded forms $300 + 40 + 2$ and $300 + 20 + 4$.
Solution: The hundreds parts match (300 and 300). The tens parts decide: 40 is greater than 20, and the ones cannot overturn a tens difference. $300 + 40 + 2 > 300 + 20 + 4$
- Maya read 246 pages; Leo read 264 pages. Compare, and who read more?
Solution: Hundreds tie (2 and 2); compare tens: 4 tens versus 6 tens, so $246 < 264$

The greater count, 264, belongs to Leo, so *Leo* read more pages.

5. Order least to greatest: 512, 251, 521

Solution: Compare hundreds first: 251 has only 2 hundreds, so it is least. 512 and 521 tie at 5 hundreds; tens decide: 1 ten beats nothing — 512 has 1 ten, 521 has 2 tens, so 512 comes before 521.

251, 512, 521

6. Green Hill 730 m, Stone Hill 703 m: compare the heights.

Solution: Hundreds tie (7 and 7). Tens decide: 730 has 3 tens while 703 has 0 tens — the zero holds the tens place. So Green Hill is taller:

$730 > 703$

7. Greatest three-digit number from the digits 2, 5, 8 (used once each).

Solution: The biggest digit must take the biggest place: 8 in the hundreds, then 5 in the tens, then 2 in the ones: $800 + 50 + 2$.

852

8. Order the can counts 318, 381, 313 from least to greatest.

Solution: All three have 3 hundreds, so compare tens: 313 and 318 have 1 ten; 381 has 8 tens, so 381 is greatest. Between 313 and 318 the ones decide: 3 ones are fewer than 8 ones.

313, 318, 381

9. Sam says 476 is greater than 512 because 7 tens beat 1 ten. True comparison, and why is he wrong?

Solution: Compare hundreds *before* tens: 476 has 4 hundreds, 512 has 5 hundreds, so

$476 < 512$

What a good explanation contains: the hundreds place is compared first, and 5 hundreds beat 4 hundreds no matter what the tens digits are; tens only matter when the hundreds tie.

10. Challenge: order 907, 790, 970, 709 from least to greatest.

Solution: Compare hundreds: 709 and 790 have 7 hundreds; 907 and 970 have 9 hundreds, so the 7-hundreds pair comes first. Within each pair the tens decide: 709 (0 tens) before 790 (9 tens), and 907 (0 tens) before 970 (7 tens).

709, 790, 907, 970